



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

Symbiosis Institute of Technology, Pune

Faculty of Engineering

CSE- Academic Year 2026-27

Java Programming – Lab Batch 2025-29

Lab Assignment No:- 21	
Name of Student	RAM SHARMA
PRN No.	25070122150
Batch	CSE-B2
Class	PROGRAMMING WITH JAVA
Academic Year & Semester	2nd-year and 3rd-semester
Date of Submission	04/10/2026
Title of Assignment:	Demonstration and exploration of JDBC classes and interfaces (DriverManager,Connection, Statement).
Problem Statement	Demonstration and exploration of JDBC classes and interfaces (DriverManager,Connection, Statement).
Learning objective	To understand and use the core JDBC classes for database connectivity.
Theory	The JDBC API consists of several classes and interfaces that facilitate database connectivity. The DriverManager class is responsible for loading the appropriate JDBC driver and establishing a connection with the database using a database URL, username, and password. The Connection interface represents an active session between the Java application and the database server. It provides methods to create SQL statements and manage transactions. The Statement interface executes static SQL queries and updates.

	These classes work together to establish communication between Java programs and databases. Understanding these core components is essential for developing database applications using Java.
Programme	1. Create a program to connect Java with database and display database connection status. <pre> =>import java.sql.Connection; import java.sql.DriverManager; import java.sql.Statement; public class E1 { public static void main(String[] args) { String url = "jdbc:mysql://localhost:3306/jdbc_demo"; String user = "root"; String password = "YOUR_MYSQL_PASSWORD"; try { // Load JDBC Driver Class.forName("com.mysql.cj.jdbc.Driver"); // Establish connection using DriverManager Connection con = DriverManager.getConnection(url, user, password); System.out.println("Database connected successfully."); // Create Statement Statement stmt = con.createStatement(); System.out.println("Statement object created successfully."); // Close connection con.close(); } catch (Exception e) { System.out.println("Error: " + e.getMessage()); } } } </pre>

```
}
```

2. Create a Java program to connect with a database and display the **connection status**.After successful connection, show a message indicating that the **Student database is connected successfully**

```
=>import java.sql.Connection;  
import java.sql.DriverManager;  
import java.sql.Statement;
```

```
public class E2 {
```

```
    public static void main(String[] args) {
```

```
        String url = "jdbc:mysql://localhost:3306/jdbc_demo";  
        String user = "root";  
        String password = "YOUR_MYSQL_PASSWORD";
```

```
        try {
```

```
            // Load JDBC Driver  
            Class.forName("com.mysql.cj.jdbc.Driver");
```

```
            // Connect to database  
            Connection con = DriverManager.getConnection(  
                url, user, password  
            );
```

```
            System.out.println("Student database connected successfully.");
```

```
            // Create Statement  
            Statement stmt = con.createStatement();
```

```
            System.out.println("Statement object created successfully.");
```

```
            // Close connection  
            con.close();
```

```
        } catch (Exception e) {
```

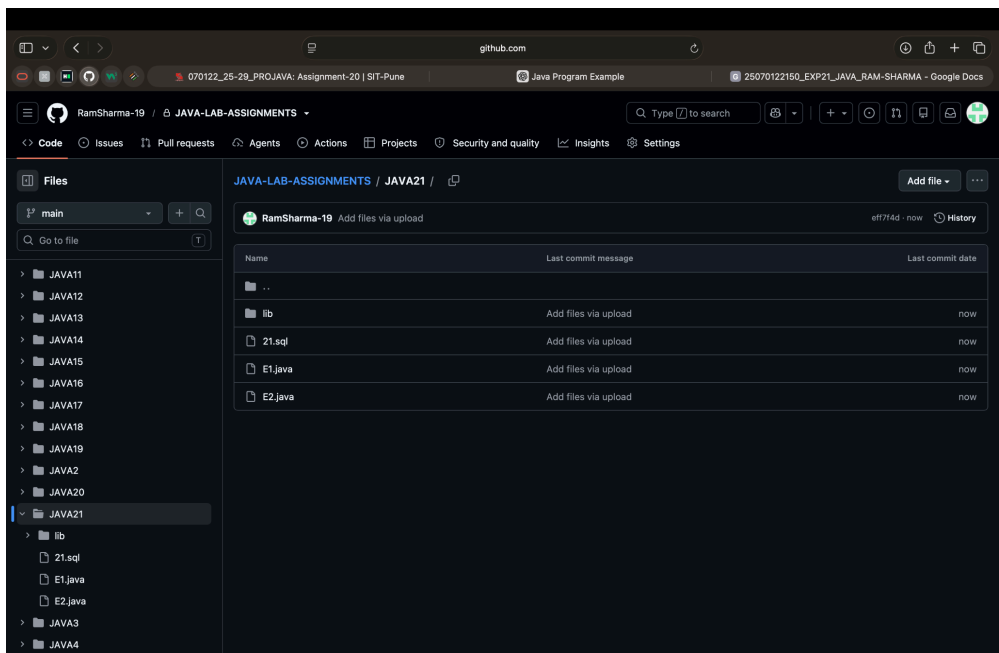
```
            System.out.println("Error: " + e.getMessage());
```

```
        }
```

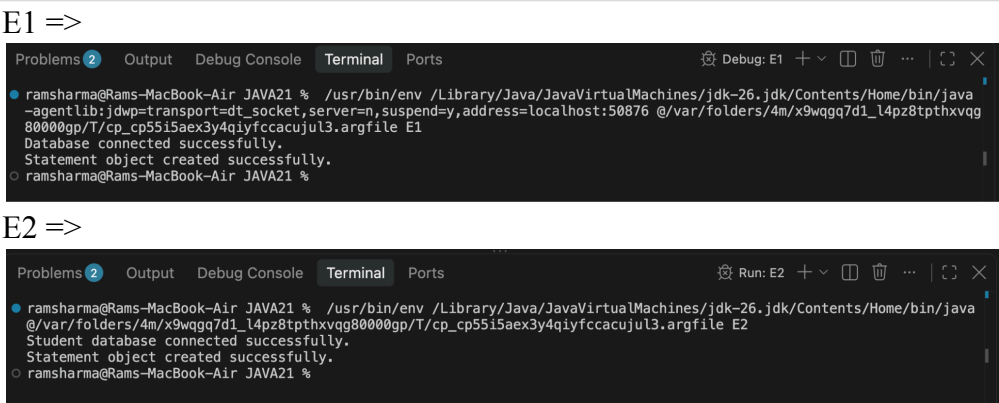
```
    }
```

```
}
```

SQL:



Output
Screenshots
(if applicable)



Github
Screenshot
Conclusion

In this assignment, the basic JDBC classes **DriverManager**, **Connection**, and **Statement** were explored. The Java programs successfully established a connection between the Java application and MySQL database. The Connection object was used to maintain communication with the database, while the Statement object was created for executing SQL statements. The connection status was displayed after successful connection. This experiment helped in understanding the basic components required for database connectivity using JDBC.

